

Roberta Cimorelli Belfiore

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Education and Training

University of Molise — **Pesche, Italy**
PhD in Computer Science — *2023–Current*
Supervisor: Prof. Anna Lisa Ferrara
PhD Program in Biology and Applied Sciences, Applied Sciences curriculum

University of Molise — **Pesche, Italy**
Master's Degree in Software Systems Security — *2021–2023*
Cum Laude
Thesis: *Identity-Based Matchmaking Encryption: A Generic Construction and Instantiations from Standard Lattice Assumptions*

University of Molise — **Pesche, Italy**
Bachelor's Degree in Computer Science — *2019–2021*

Visiting Research Experience

Fondazione Bruno Kessler (FBK) — **Trento, Italy**
PhD Visiting Researcher — *Sep 2025 – Mar 2026*
Research stay at the Center for Cybersecurity of FBK under the supervision of Prof. Silvio Ranise. The research focused on applied cryptography and access control for protected data, with particular emphasis on analyzing the Access Control Encryption (ACE) primitive and its potential synergies with alternative cryptographic access control approaches developed by the host institution, and on the implementation of the Contextual Reading ACE framework.

Newcastle University — CryptoLab — **Newcastle, UK**
PhD Visiting Researcher — *Feb 2025 – Aug 2025*
Research within the CryptoLab at the School of Computing, supervised by Dr. Essam Ghadafi, on the design of cryptographic primitives for access control over encrypted data, with a focus on efficient post-quantum constructions.

Newcastle University — **Newcastle, UK**
Short-term Visit — *Jul 2024*
Research discussions on collaborative opportunities and invited talk (see Schools, Conferences and Seminars).

Research Interests

My research combines theoretical cryptographic foundations with practical applications, focusing on cryptographic access control, privacy-preserving authentication, forward security and quantum-resilient protocols. My recent work has explored variants of cryptographic primitives such as Identity-Based Matchmaking Encryption, Access Control Encryption, Hierarchical Key Assignment Schemes and Sanitizable

Signatures, aiming to strengthen privacy, accountability and resilience to key compromise in real-world systems.

Teaching and Mentoring

University of Molise, Department of Biosciences and Territory — **Pesche, Italy**
Subject Expert (Cultore della Materia) — *2023–Present*

- Member of examination boards and support to teaching activities.
- Courses of Prof. Gennaro Parlato (B.Sc. Computer Science):
  - *Algorithms and Data Structures (with Laboratory)*
  - *Theory of Computing*
  - *Program Analysis*
- Courses of Prof. Anna Lisa Ferrara:
  - *Computer Networks* (B.Sc. Computer Science)
  - *Logic and Foundations of Computer Science* (B.Sc. Computer Science)
  - *Cryptography* (M.Sc. Software Systems Security)
  - *Networking Security and Software Security* (M.Sc. Software Systems Security)
- Courses of Prof. Manuela Flores (B.Sc. Computer Science):
  - *Object-Oriented Programming*

University of Molise — **Isernia, Italy**
Orientation and Tutoring Assistant — *Feb 2023 – May 2023*

Selected through the university call for orientation and tutoring tutors.

- Ongoing tutoring and academic support for bachelor's and master's students.
- Dedicated tutoring for working students and for students with learning disabilities (D.S.A.).
- Support to students with administrative tasks related to their degree programs.

Thesis Co-supervision

2025/26: **Giada Cancelliere** – B.Sc. Computer Science
Analisi di Ridondanza in Firme Digitali Avanzate su Gruppi Bilineari

2024/25: **Rossella Mastroiacovo** – B.Sc. Computer Science
Indicizzazione multidimensionale su dati cifrati: un approccio per dataset dinamici

2024/25: **Michela Martino** – B.Sc. Computer Science
Un'analisi comparativa dei modelli di minaccia nei protocolli AKE, PAKE e A-AKE

2023/24: **Alessio Del Riccio** – M.Sc. Software Systems Security
Revisiting SAE+: Towards Post-Server Compromise Security

2023/24: **Federica Patriarca** – M.Sc. Software Systems Security
Studio della sicurezza dei protocolli di autenticazione Wi-Fi: da WPA2 a SAE+

2022/23: **Gabriella Testa** – M.Sc. Software Systems Security
Analyzing Access Control Policies from Smart Homes to Smart Buildings

Academic Service

Program Committees

2026: 31st European Symposium on Research in Computer Security (ESORICS 2026)

Organizing Committees

2026: Web & Communications Support, 1st International Summer School: Biosciences, Informatics, Tourism, Engineering, Sustainability (BITES 2026)

2024: Web Chair, 38th IFIP WG 11.3 Conference on Data and Applications Security and Privacy (DBSec 2024)

Journal Reviewing

IEEE Transactions on Dependable and Secure Computing (TDSC)

ACM Transactions on Internet Technology (TOIT)

Journal of Information Security and Applications (JISA)

External Reviewing

2024: Sub-reviewer, 38th IFIP WG 11.3 DBSec 2024

Schools, Conferences and Seminars

Conference Talk: *Dual-Forward-Secure Sanitizable Signatures*
31st European Symposium on Research in Computer Security (ESORICS '26), Rome, Italy, 14–18 Sep 2026.

Summer School: Participation at the *1st International Summer School: Biosciences, Informatics, Tourism, Engineering, Sustainability (BITES 2026)*, Termoli, Italy, 25–28 Aug 2026.

Conference: 45th Annual International Conference on the Theory and Applications of Cryptographic Techniques (EUROCRYPT '26), Rome, Italy, 10–14 May 2026.

PhD School: Participation at the *CyberDI 2025 – Advanced Techniques in Digital Identity PhD School*, Tropea, Italy, 20–23 Oct 2025.

Invited Talk: *Analyzing Access Control Policies in Smart Environments*
Newcastle University, Newcastle upon Tyne, UK, 25 Jul 2024.

Conference Talk: *Identity-Based Matchmaking Encryption from Standard Lattice Assumptions*
22nd International Conference on Applied Cryptography and Network Security (ACNS '24), Abu Dhabi, UAE, 05–08 Mar 2024.

Conference Talk: *Security Analysis of Access Control Policies for Smart Homes*
28th ACM Symposium on Access Control Models and Technologies (SACMAT '23), Trento, Italy, 07–09 Jun 2023.

Outreach

Poster *Cryptographic Solutions for Anonymity and Privacy in the Post-Quantum Era*, Giornata della Ricerca, University of Molise.

Participation in the PhD EXPO (2nd and 3rd editions), Campobasso.

Participation in the Notte della Ricerca (European Researchers' Night), MUSE – Science Museum, Trento.

Awards

Sep 2026: **Student Travel Grant**, 31st European Symposium on Research in Computer Security (ESORICS 2026)

Jul 2026: **Best Paper Award**, DBSec 2026, for the paper *Anonymous Hierarchical Key Assignment Schemes*

Oct 2025: **Mobility Grant**, CyberDI 2025 – Advanced Techniques in Digital Identity PhD School

Feb 2025: **"Prof. Mario Massimo Petrone" Thesis Award**, for the most meritorious Master's thesis in Software Systems Security (LM-66), A.Y. 2023–2024

Sep 2024: **CLUSIT Thesis Award**, 19th edition

Jan 2024: **Student Travel Grant**, 22nd International Conference on Applied Cryptography and Network Security (ACNS 2024)

Funded Research Projects

Verifica di proprietà di sicurezza nello sviluppo del software
Team member of the research project coordinated by Prof. Anna Lisa Ferrara, as part of the Departmental Research Projects – Start-up 2023 Call of the University of Molise (UNIMOL).

INdAM-GNCS Project 2025 – CUP_E53C24001950001
Team member of the research project coordinated by Prof. Manuela Flores, funded by INdAM-GNCS.

Research Groups

2023–Now: **Program Analysis in the Clouds Lab (PAC Lab)**, University of Molise
Member of the research group coordinated by Prof. Gennaro Parlato and Prof. Anna Lisa Ferrara.

2024–Now: **INdAM-GNCS**, UNIMOL Research Unit
Member of the research group of the Istituto Nazionale di Alta Matematica "Francesco Severi" (INdAM), UNIMOL research unit, administratively based at the Department of Biosciences and Territory.

2025–2026: **Center for Cybersecurity**, Fondazione Bruno Kessler, Trento
Member of the research group led by Prof. Silvio Ranise (see Visiting Research Experience).

2025: **CryptoLab**, School of Computing, Newcastle University
Member of the research group led by Dr. Essam Ghadafi (see Visiting Research Experience).

Languages

Italian: Native

English: B2 – English Speaking Board (ESB) certificate, 2022

Publications

Peer-reviewed Conference Papers

[1]: **Cimorelli Belfiore, R.**, & Ferrara, A. L. (2026). *Dual-Forward-Secure Sanitizable Signatures*. In Spognardi, A., Sun, K., De Gaspari, F., & Mancini, L. V. (Eds.), Computer Security – ESORICS 2026, Lecture Notes in Computer Science, vol. 16971, pp. 315–335, Springer.

[2]: **Cimorelli Belfiore, R.**, De Santis, A., Ferrara, A. L., Flores, M., & Masucci, B. (2026). *Anonymous Hierarchical Key Assignment Schemes*. In Palanisamy, B., & Samarati, P. (Eds.), Data and Applications Security and Privacy XL (DBSec 2026), Lecture Notes in Computer Science, vol. 16576, pp. 224–243, Springer. **Best Paper Award.**

[3]: **Cimorelli Belfiore, R.**, Ferrara, A. L., & Masucci, B. (2026). *A Framework for Context-Aware Read Authorization over Encrypted Data*. In Proceedings of the 23rd International Conference on Security and Cryptography (SECRYPT 2026), pp. 121–132, SciTePress.

[4]: **Cimorelli Belfiore, R.**, De Santis, A., Ferrara, A. L., & Masucci, B. (2024). *Hierarchical Key Assignment Schemes with Key Rotation*. In Proceedings of the 29th ACM Symposium on Access Control Models and Technologies (SACMAT 2024), ACM.

[5]: **Cimorelli Belfiore, R.**, De Cosmo, A., & Ferrara, A. L. (2024). *Identity-Based Matchmaking Encryption from Standard Lattice Assumptions*. In Pöpper, C., & Batina, L. (Eds.), Applied Cryptography and Network Security (ACNS 2024), Lecture Notes in Computer Science, vol. 14584, Springer.

[6]: **Cimorelli Belfiore, R.**, & Ferrara, A. L. (2023). *Security Analysis of Access Control Policies for Smart Homes*. In Proceedings of the 28th ACM Symposium on Access Control Models and Technologies (SACMAT 2023), ACM.

Submitted Papers and Preprints

[7]: **Cimorelli Belfiore, R.**, De Santis, A., Ferrara, A. L., & Masucci, B. (2025). *Hierarchical Key Assignment Schemes with Key Rotation and Bidirectional Secret Derivation*. Under revision at ACM Transactions on Internet Technology.

[8]: **Cimorelli Belfiore, R.**, Ferrara, A. L., & Ghadafi, E. (2026). *Strengthening Access Control Encryption: Unique Attribution and Misattribution Resistance*. Submitted to an international conference.

[9]: **Cimorelli Belfiore, R.**, Ferrara, A. L., & Ghadafi, E. (2025). *Sender-Authenticated ACE: Bridging Practicality and Efficiency*. Preprint.